

Neural Network Project Report

MNIST Handwritten Digit Classification

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Course: [Neural Network]

1. Introduction

The objective of this project is to build a neural network model capable of recognizing handwritten digits. The problem is a classification task where the model predicts digits from 0 to 9 based on input images. A Multilayer Perceptron (MLP) was selected to solve this problem due to its simplicity, efficiency, and effectiveness in handling structured input data like flattened images.

2. Dataset and Preprocessing

The dataset used in this project is the MNIST dataset, a standard benchmark in machine learning for image classification tasks. The dataset is directly loaded using TensorFlow/Keras and consists of:

- 60,000 training images.
- 10,000 testing images.
- Images have a resolution of 28×28 pixels in grayscale.

Before training the model, the following preprocessing steps were applied to the data:

- **Normalization:** Pixel values were scaled from the range [0, 255] down to [0, 1] to ensure numerical stability during gradient descent.
- **Flattening:** Each 28×28 image was reshaped into a 1D vector of size 784 to match the input requirements of the MLP.
- **One-hot encoding:** Target labels were converted into a categorical format to align with the multi-class classification objective.

3. Model Architecture and Experimental Setup

Two different MLP architectures were implemented and evaluated to compare performance. The models were trained using the Adam optimizer and the Categorical Crossentropy loss function over 10 epochs, with a batch size of 64 and a 20% validation split.

Layer / Parameter	Experiment 1 Architecture	Experiment 2
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		Architecture
Input Layer	784 neurons	784 neurons
Hidden Layer 1	128 neurons (ReLU)	256 neurons (ReLU)
Dropout Layer	Rate: 0.2	Rate: 0.3
Hidden Layer 2	64 neurons (ReLU)	128 neurons (ReLU)
Output Layer	10 neurons (Softmax)	10 neurons (Softmax)

4. Results

The performance of both models after 10 epochs is summarized in the table below. Experiment 2 demonstrated a slight but noticeable improvement over Experiment 1.

Model	Final Loss	Final Accuracy
Experiment 1	0.0850	97.68%
Experiment 2	0.0824	97.78%

5. Visualizations

The training and validation curves for both loss and accuracy are plotted below. These graphs clearly show that the models converged properly and did not suffer from significant overfitting, thanks to the implementation of dropout layers.

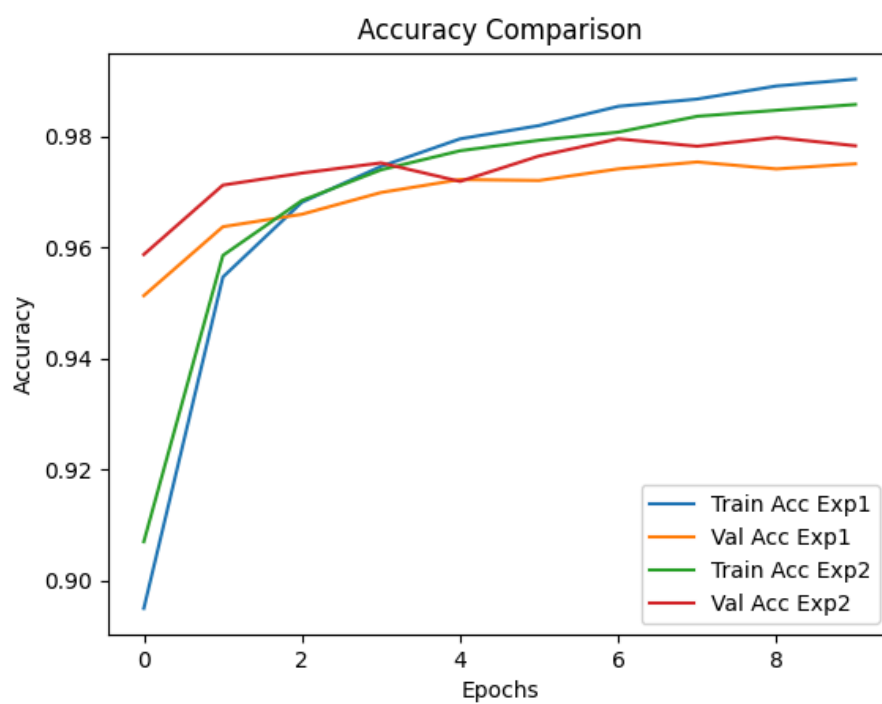


Figure 1: Training and Validation Accuracy over Epochs

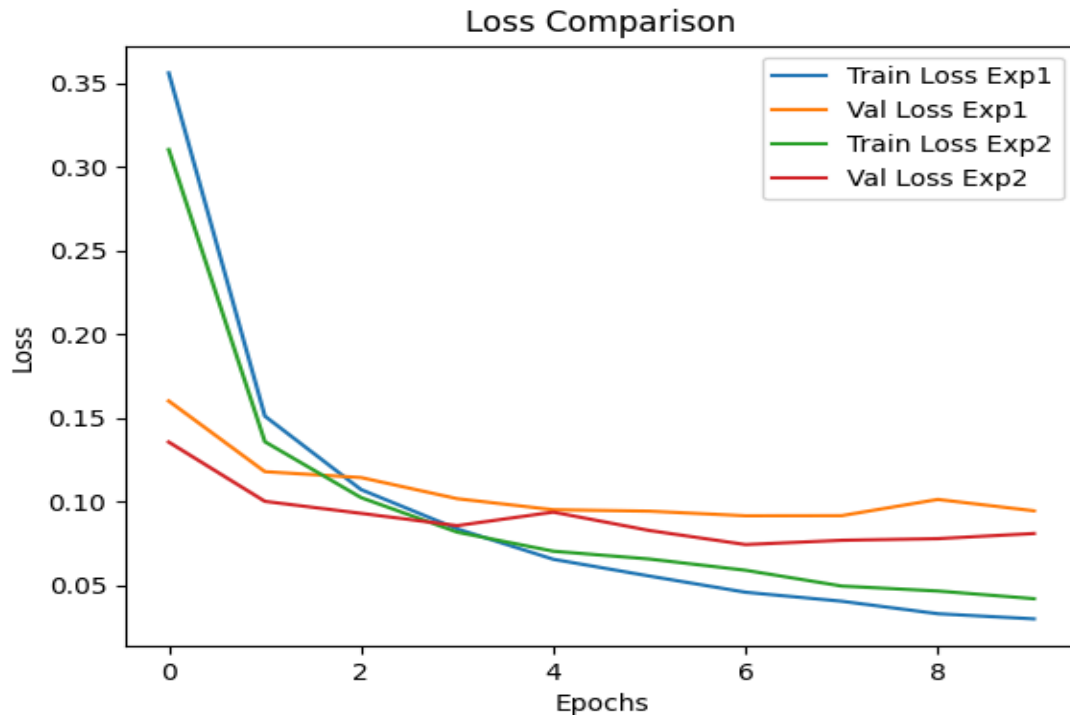


Figure 2: Training and Validation Loss over Epochs

6. Discussion and Conclusion

The experimental results show that increasing the number of neurons enhances the model's ability to capture more complex patterns in the input data. In addition, incorporating dropout in the second model helped reduce overfitting and improved generalization on unseen test data.

Although the improvement in accuracy between the two experiments is relatively small (97.68% vs 97.78%), the second model consistently achieved better performance with both higher accuracy and lower loss. This indicates that combining a larger network with appropriate regularization leads to more reliable results.

In conclusion, a Multilayer Perceptron (MLP) was successfully implemented to classify handwritten digits from the MNIST dataset with high accuracy exceeding 97%. The results confirm that careful selection of model architecture and the use of regularization techniques,

such as dropout, can significantly improve the performance and stability of neural network models in image classification tasks.